Day 4 of 15 · Student Activity Sheets

Day 4 Activities — Data Sources, Databases and SQL Retrieval

TESDA Data Analytics Level III · Batch DAL3-2026

Name: _____ Date: _____ Workstation #: _____

Folder you need: `Day04/assets/Day04_SQL_Lab/` (contains `aling_nena.db` and `Day04_Queries.sql`)

Before you begin:

1. Copy the whole `Day04_SQL_Lab` folder into `DAL3/Day04/`. Rename the script to `Day04_<Surname>.sql` (e.g. `Day04_Santos.sql`).
2. In VS Code: **Ctrl + K, Ctrl + O** → choose your `Day04_SQL_Lab` folder → **Select Folder**. Open your `.sql` file from the Explorer.
3. Type each answer **on the empty line under its question** in the script. End every query with `;`.
4. Run a query: **select it** → **Ctrl + E, Ctrl + E**. Write the **row count** (bottom-right of the result panel) on this sheet.
5. Save often: **Ctrl + S**. Stuck? Re-watch that chapter of the demo video. The PRESS box shows every key.

Activity	Title	Mode	Time	Points
4.0	Warm-up: Sort the Data Sources	Individual	15 min	10
4.1	Explore the Aling Nena Database	Individual	25 min	15
4.2	SELECT and DISTINCT (Q1–Q4)	Individual	20 min	15
4.3	Filtering with WHERE (Q5–Q11)	Individual	45 min	30
4.4	Sorting and Top-N (Q12–Q15)	Pair	30 min	20
4.5	Data Source Inventory for Aling Nena	Group of 3–4	25 min	10
	Total		**≈ 2 hrs 40 min**	**100**

Activity 4.0 — Warm-up: Sort the Data Sources (10 pts)

No computer. For each data source, tick **Internal** or **External**, and write **S** (structured), **SS** (semi-structured) or **U** (unstructured).

#	Data source	Internal	External	S / SS / U
1	POS sales receipts saved in the store's database			
2	PAGASA daily temperature forecast table			

#	Data source	Internal	External	S / SS / U
3	Comments on the store's Facebook page			
4	Suki membership list in an Excel file			
5	Photos of delivery receipts from the supplier			
6	GCash settlement report downloaded as CSV			
7	Voice messages from customers ordering by Messenger			
8	Order data sent by a delivery app as JSON			
9	PSA monthly inflation rates (table on the website)			
10	CCTV video from the Pasig branch			

Activity 4.1 — Explore the Aling Nena Database (15 pts)

1. Click the **SQLTools** icon (database cylinder) in the left bar. Expand **Aling Nena** → **Tables**.
2. Expand each table to see its columns. Right-click a table → **Show Table Records** to peek at the rows.
3. Count the rows of each table. Type and run this in your script (we'll learn COUNT properly tomorrow):

```
SELECT COUNT(*) FROM branches;    -- then change branches to products, customers, sales
```

Table	Number of rows	Primary key	Foreign key(s), and which table each points to
branches			
products			
customers			
sales			

Questions

a) Sale **TXN-10003** has an empty (NULL) `customer_id`. What kind of customer was it? Why is that not an error?

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b) Why does the `sales` table store `product_id` (e.g. 107) and not the full product name?

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c) Which branch has status **Opening Soon**? _____ Do you expect it to have any sales? _____

Activity 4.2 — SELECT and DISTINCT (15 pts)

Write each query in your script under its question, run it, and record the row count.

Q	Task	Rows
Q1	Show **every column** of every branch.	
Q2	Show only the **product name, category and unit price** of every product.	
Q3	Show the **first name, last name and city** of every customer.	
Q4	List **each product category once** (no repeats).	

Write your Q4 query here:

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d) In Q2, what happens if you forget the comma between `category` and `unit_price`? Try it, then explain. *(Hint: SQL thinks `unit_price` is a new name for `category`.)*

Activity 4.3 — Filtering with WHERE (30 pts)

Q	Task	Rows
Q5	Name and price of all products in the **Beverages** category.	
Q6	Name and price of products that cost **more than ₱60**.	
Q7	Sale ID, date and total amount of **Pasig (BR03)** sales paid with **GCash**.	
Q8	Sale ID, date and total amount of sales from **1 to 7 February 2026** (both days included).	
Q9	Name and price of every product with the word **Milk** in its name.	
Q10	ID, first name and last name of customers who have **no email** address.	
Q11	Sale ID, payment method and quantity of **e-wallet** sales (**GCash or Maya**) with a quantity of **3 or more**.	

Write your Q8 and Q11 queries here:

Questions

e) Run `SELECT * FROM customers WHERE email = NULL;`. How many rows? _____ Why? What should you write instead?

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f) Q11 can be written with `IN` or with `OR`. Write the `OR` version. Be careful with **AND before OR**: you need parentheses.

Activity 4.4 — Sorting and Top-N (Pair, 20 pts)

*Pair work: one person types (the **driver**), the other checks the spelling and the results (the **navigator**). Switch after Q13.*

Q	Task	Rows
Q12	The **5 most expensive** products (name and price), most expensive first.	
Q13	The **10 biggest sales** (sale ID, branch ID, total amount), biggest first. Ties: smaller sale ID first.	
Q14	Each **payment method** used in the sales table **once**, in **A to Z** order.	
Q15	Sale ID, time and total amount of all **Valentine's Day** sales (**14 February 2026**), earliest first.	

Questions

g) What is the biggest single sale amount? ₱ _____ How many sales tie at that amount? _____ Which branch had most of the top 10? _____

h) Q15 gives very few rows for Valentine's Day. Give one possible reason, and one more data source Aling Nena could check. *(Integrated: Use information systematically.)*

Activity 4.5 — Data Source Inventory for Aling Nena (Group, 10 pts)

(Integrated: Use information systematically.) Aling Nena asks: **"Why are my sales lower on rainy days, and what should I stock?"**
As a team, list **five data sources** you would use to answer her. At least one must be external, and at least one must come from `aling_nena.db`.

#	Data source	Internal / External	S / SS / U	Who owns it / how to get it	Quality concern (complete? up to date?)
1					
2					
3					
4					
5					

Rubric: five realistic sources (4), correct classification (3), sensible quality concern (3). The reporter presents in 2 minutes.

Daily Reflection Log

1. Explain *primary key* to a friend, in your own words:

   _____

2. The SQL mistake I made most today, and how I fixed it:

   _____

3. One question about my own work or barangay that I could answer with a database:

   _____

Submit: Day04_<Surname>.sql (all 15 queries, each runs without error) and this sheet.